Quiz 1.1

Name and Uniqname:

Each question has **one correct answer.** Circle the correct answer only.

Question 1

Which R expression is a Monte Carlo estimate of $\int_0^1 \cos(\pi x/2)\,dx$?

1. `cos(pi * mean(runif(1000)) / 2)`
2. `mean(cos(pi * runif(1000) / 2))`
3. `mean(cos(pi * (1:1000) / 2000))`
4. `mean(runif(1000)) * cos(pi / 2)`

Question 2

You want to estimate $\int_0^\infty g(x)\,dx$ using draws `x <- rexp(n)` from the Exp(1) distribution. Which estimator is correct?

1. `mean(g(x) * dexp(x))`
2. `mean(g(x))`
3. `mean(g(dexp(x)))`
4. `mean(g(x) / dexp(x))`

Question 3

Let `x <- rexp(10000)`. What does `mean(x <= 2)` estimate?

1. $P(X \leq 2)$
2. $\mathrm{E}\left(\min(X, 2)\right)$
3. The density of X at 2
4. $\mathrm{E}\left(X\right) \times P(X \leq 2)$

Question 4

You estimate $\theta = P(X > 3)$ by the sample mean of the indicators $\mathbf{1}(X_i > 3)$. Why is `binom.test` preferred over `t.test` for the confidence interval?

1. `t.test` cannot handle values of 0 and 1.
2. `binom.test` intervals are always narrower.
3. The indicators are Bernoulli, so a proportion interval applies exactly.
4. The indicators are not identically distributed.

Question 5

With n function evaluations in k dimensions, grid-based integration has error rate $O(n^{-2/k})$, while Monte Carlo has error rate $O(n^{-1/2})$.

For an integral with 8 dimensions ($k = 8$), which method would be expected to have less error for a very large number of function evaluations (i.e. n is large)?

1. Grid-based
2. Monte Carlo
3. They would be about the same
4. It depends on the function being integrated, not k or n.